1. Two-Dimensional Problems in FEM

1.1 Introduction

- Many real problems (thin plates, dams, retaining walls, gears, tunnels) can be idealised as 2-D problems – plane stress or plane strain.
- Each node has 2 dof: displacements u (x-direction) and v (y-direction).
- The simplest 2-D element is the triangle, which fits any boundary shape, so it is the most popular one.
- Two types of triangular elements: CST (Constant Strain Triangle, 3 nodes) and LST (Linear Strain Triangle, 6 nodes).

1.2 General procedure for 2-D analysis

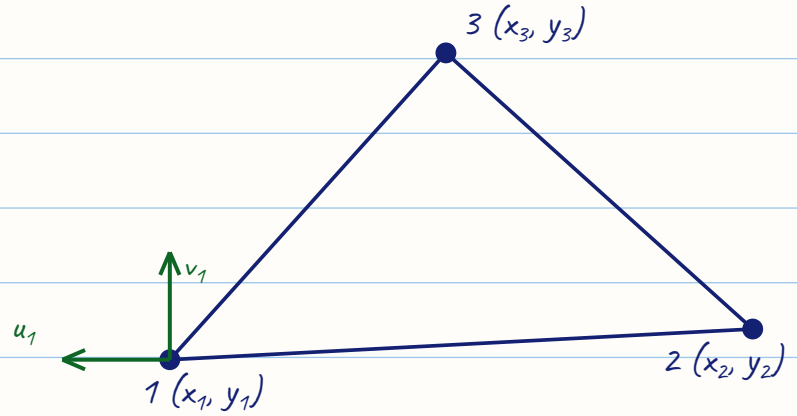
- Divide the body into triangular (or quadrilateral) elements; number nodes and elements.
- Assume displacement functions u(x,y), v(x,y) in each element.
- Find [B], [D] and the element stiffness matrix [k].
- Assemble [K], apply loads and boundary conditions, solve [K]{U} = {F}.
- Back-calculate strains and stresses; plot the stress contours.

Rule: put a fine mesh at stress concentrations (holes, notches, corners); avoid very thin, long triangles.

2. Constant Strain Triangle (CST) Element

2.1 Description

- Three nodes (1, 2, 3) at the corners, numbered anticlockwise. Coordinates (x_i, y_i).
- Six dof per element: $\{q\} = \{u_1\ v_1\ u_2\ v_2\ u_3\ v_3\}^T$.



CST: 3 nodes, 6 dof

$u = a_1 + a_2x + a_3y$

$v = a_4 + a_5x + a_6y$

ε = constant over element

2.2 Displacement function

$$u = a_1 + a_2x + a_3y$$

$$v = a_4 + a_5x + a_6y$$

- The six constants are found from the six nodal displacements. For u:

$$u_1 = a_1 + a_2x_1 + a_3y_1$$

$$u_2 = a_1 + a_2x_2 + a_3y_2$$

$$u_3 = a_1 + a_2x_3 + a_3y_3$$

- Solving by Cramer's rule gives a_1, a_2, a_3 in terms of u_1, u_2, u_3

(similarly a_4, a_5, a_6 in terms of v's).

2.3 Shape functions

$u = N_1u_1 + N_2u_2 + N_3u_3$, $v = N_1v_1 + N_2v_2 + N_3v_3$

$N_i = (\alpha_i + \beta_i x + \gamma_i y) / 2A$, $i = 1, 2, 3$

$\alpha_1 = x_2y_3 - x_3y_2$, $\beta_1 = y_2 - y_3$, $\gamma_1 = x_3 - x_2$

$\alpha_2 = x_3y_1 - x_1y_3$, $\beta_2 = y_3 - y_1$, $\gamma_2 = x_1 - x_3$

$\alpha_3 = x_1y_2 - x_2y_1$, $\beta_3 = y_1 - y_2$, $\gamma_3 = x_2 - x_1$

Area of triangle: $2A = x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)$

- Check: $N_1 + N_2 + N_3 = 1$ for every point; in area coordinates $N_i = L_i$.
- In matrix form $\{u\ v\}^T = [N]\{q\}$ with $[N] = [N_1\ 0\ N_2\ 0\ N_3\ 0\ ;\ 0\ N_1\ 0\ N_2\ 0\ N_3]$.

3. Strain – Displacement Matrix of CST

$\varepsilon_x = \partial u/\partial x = (\beta_1u_1 + \beta_2u_2 + \beta_3u_3) / 2A$

$\varepsilon_y = \partial v/\partial y = (\gamma_1v_1 + \gamma_2v_2 + \gamma_3v_3) / 2A$

$\gamma_{xy} = \partial u/\partial y + \partial v/\partial x = (\gamma_1u_1 + \beta_1v_1 + \gamma_2u_2 + \beta_2v_2 + \gamma_3u_3 + \beta_3v_3) / 2A$

$\{\varepsilon\} = [B]\{q\}$

$[B] = (1/2A)\ [\beta_1\ 0\ \beta_2\ 0\ \beta_3\ 0\ ;\ 0\ \gamma_1\ 0\ \gamma_2\ 0\ \gamma_3\ ;\ \gamma_1\ \beta_1\ \gamma_2\ \beta_2\ \gamma_3\ \beta_3]$

- [B] contains only nodal coordinates (no x or y) – so strains are

CONSTANT throughout the element. That is why it is called the constant strain triangle.

- Because stress = [D][B]{q}, stress is also constant in the element; the value corresponds to the centroid.

4. Stress – Strain Relation and Stiffness Matrix of CST

4.1 Stress

$$\{\sigma\} = [D]\{\varepsilon\} = [D][B]\{q\}$$

Plane stress: $[D] = E/(1-\nu^2)\ [1\ \nu\ 0\ ;\ \nu\ 1\ 0\ ;\ 0\ 0\ (1-\nu)/2]$

Plane strain: $[D] = E/[(1+\nu)(1-2\nu)]\ [1-\nu\ \nu\ 0\ ;\ \nu\ 1-\nu\ 0\ ;\ 0\ 0\ (1-2\nu)/2]$

4.2 Element stiffness matrix

- From the minimum potential energy principle:

$$[k] = \int_V [B]^T [D][B]\, dV$$

- [B] and [D] are constants, so for an element of uniform thickness t:

$$[k] = t\, A\, [B]^T\, [D]\, [B]$$

- The matrix is 6×6, symmetric, and singular (rigid body modes), with positive diagonal terms.

4.3 Nodal load vectors

- Body force (self weight) per unit volume $\{X\ Y\}$:

  $\{f_b\} = (t\,A/3)\,\{X\ Y\ X\ Y\ X\ Y\}^T$

- Traction on an edge of length l_{12} (uniform pressure p_x, p_y):

  $\{f_T\} = (t\,l_{12}/2)\,\{p_x\ p_y\ p_x\ p_y\ 0\ 0\}^T$

- Initial strain / temperature: $\{f_0\} = t\,A\,[B]^T[D]\{\varepsilon_0\}$, $\{\varepsilon_0\} = \{\alpha\Delta T\ \alpha\Delta T\ 0\}^T$ (plane stress).

4.4 Properties of CST

- Simple and easy; fits irregular boundaries; constant strain – poor accuracy where stress changes quickly, so many small elements are required.
- Displacement is continuous (linear) along each edge – compatibility is satisfied.
- Convergence is slow compared with higher order elements.

Solved example outline: given nodal coordinates, find A, β_i, γ_i, $[B]$, then $[k] = tA[B]^T[D][B]$ and $\{\sigma\} = [D][B]\{q\}$.

5. Worked Example for CST

- Problem: Triangle with nodes 1 (0, 0), 2 (3, 0), 3 (0, 2) mm, $t = 1$ mm, $E = 200$ GPa, $\nu = 0.3$, plane stress. Find $[B]$ and the area.

$$2A = 0(0-2) + 3(2-0) + 0(0-0) = 6 \rightarrow A = 3\ \text{mm}^2$$

$$\beta_1 = y_2 - y_3 = -2,\ \beta_2 = y_3 - y_1 = 2,\ \beta_3 = y_1 - y_2 = 0$$

$\gamma_1 = x_3 - x_2 = -3$, $\gamma_2 = x_1 - x_3 = 0$, $\gamma_3 = x_2 - x_1 = 3$

$[B] = (1/6)\,[-2\ 0\ 2\ 0\ 0\ 0\ ;\ 0\ -3\ 0\ 0\ 0\ 3\ ;\ -3\ -2\ 0\ 2\ 3\ 0]$

- $[D]$ for plane stress: $E/(1-\nu^2) = 200000/0.91 = 2.198 \times 10^5$ N/mm²

$[D] = 2.198\times10^5\,[1\ 0.3\ 0\ ;\ 0.3\ 1\ 0\ ;\ 0\ 0\ 0.35]$

- Suppose the nodal displacements are $u_2 = 0.001$ mm, all others zero. Then

$\varepsilon_x = \beta_2 u_2 / 2A = 2 \times 0.001 / 6 = 3.33 \times 10^{-4}$

$\varepsilon_y = 0$, $\gamma_{xy} = \gamma_2 u_2 / 2A = 0$ (since $\gamma_2 = 0$)

$\sigma_x = 2.198\times10^5 \times 3.33\times10^{-4} = 73.3$ N/mm², $\sigma_y = 0.3\,\sigma_x = 22$ N/mm², $\tau_{xy} = 0$

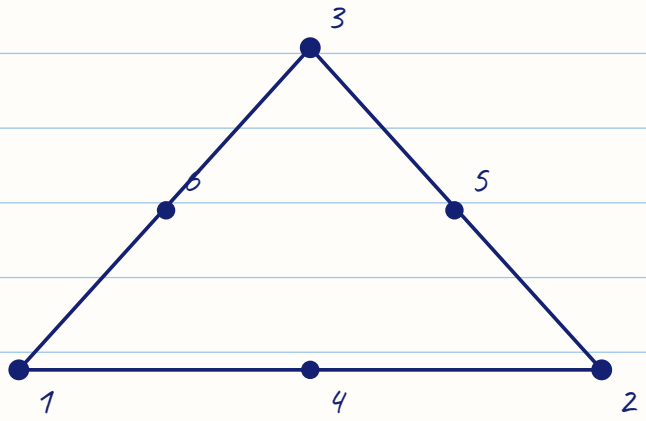
6. Linear Strain Triangle (LST) Element

6.1 Description

- Six nodes: three corner nodes and three mid-side nodes. 12 dof per element.
- Displacement is a complete quadratic polynomial:

$u = a_1 + a_2x + a_3y + a_4x^2 + a_5xy + a_6y^2$

$v = a_7 + a_8x + a_9y + a_{10}x^2 + a_{11}xy + a_{12}y^2$



LST: 6 nodes, 12 dof

$u = a_1 + a_2x + a_3y + a_4x^2 + a_5xy + a_6y^2$

strain varies linearly

6.2 Shape functions (area coordinates)

$$N_1 = L_1(2L_1 - 1),\ N_2 = L_2(2L_2 - 1),\ N_3 = L_3(2L_3 - 1)$$

$$N_4 = 4L_1L_2,\ N_5 = 4L_2L_3,\ N_6 = 4L_3L_1$$

- Sum of $N_i = 1$. Each $N_i = 1$ at its own node, 0 at others.

6.3 Strain and stress

- Strains are obtained by differentiating the shape functions; they vary LINEARLY within the element:

  $\varepsilon_x = \partial u/\partial x,\ \varepsilon_y = \partial v/\partial y,\ \gamma_{xy} = \partial u/\partial y + \partial v/\partial x$

  $[B]$ is a 3×12 matrix whose terms are linear functions of x and y.

  $[k] = t \int_A [B]^T [D] [B]\, dA$ (12 × 12)

- Integration is done using area-coordinate formula or Gauss numerical integration.
- Stress $\{\sigma\} = [D][B]\{q\}$ also varies linearly – more accurate than CST for the same number of nodes; curved boundaries can be fitted

with mid-side nodes.

6.4 CST vs LST

- CST: 3 nodes, 6 dof, constant strain, [k] 6×6, simple but needs fine mesh.
- LST: 6 nodes, 12 dof, linear strain, [k] 12×12, more accurate, more computation.
- LST gives good results with fewer elements; CST requires a large number of elements to reach the same accuracy.

7. Isoparametric Formulation

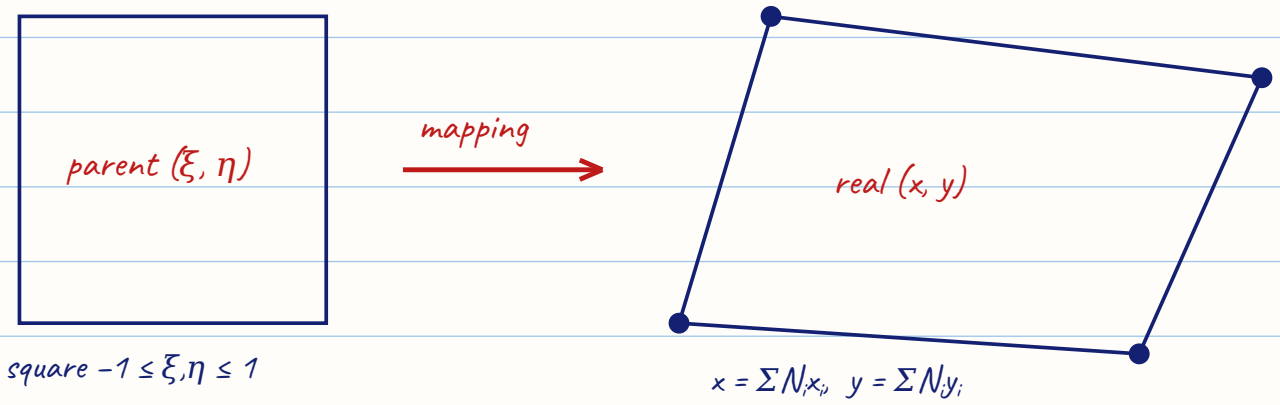
7.1 Concept

- Iso = same. In an isoparametric element, the SAME shape functions are used to define (i) the geometry (x, y) and (ii) the displacement (u, v).

$$x = \Sigma\, N_i(\xi,\eta)\, x_i,\ y = \Sigma\, N_i(\xi,\eta)\, y_i$$

$$u = \Sigma\, N_i(\xi,\eta)\, u_i,\ v = \Sigma\, N_i(\xi,\eta)\, v_i$$

- Subparametric: geometry uses lower-order functions than displacement. Superparametric: geometry uses higher order than displacement.
- A regular parent element in (ξ, η) is mapped to a distorted (curved or skew) element in (x, y).



7.2 Isoparametric quadrilateral (4 node)

$N_1 = \frac{1}{4}(1-\xi)(1-\eta)$, $N_2 = \frac{1}{4}(1+\xi)(1-\eta)$, $N_3 = \frac{1}{4}(1+\xi)(1+\eta)$, $N_4 = \frac{1}{4}(1-\xi)(1+\eta)$

7.3 Jacobian matrix

- Derivatives with respect to x, y are needed for strain, but N_i are functions of ξ, η. By chain rule:

$$\{\partial N/\partial\xi \ \partial N/\partial\eta\}^T = [J]\{\partial N/\partial x \ \partial N/\partial y\}^T$$

$$[J] = [\partial x/\partial\xi \ \partial y/\partial\xi \ ; \partial x/\partial\eta \ \partial y/\partial\eta] = [\Sigma(\partial N_i/\partial\xi)x_i \ \Sigma(\partial N_i/\partial\xi)y_i \ ; \Sigma(\partial N_i/\partial\eta)x_i \ \Sigma(\partial N_i/\partial\eta)y_i]$$

$$\{\partial N/\partial x \ \partial N/\partial y\}^T = [J]^{-1}\{\partial N/\partial\xi \ \partial N/\partial\eta\}^T$$

- Element area: $dA = dx\,dy = \det[J]\,d\xi\,d\eta$. $\det[J]$ must be positive everywhere – otherwise the element is too distorted (folded).

7.4 Strain–displacement matrix and stiffness

$\{\varepsilon\} = [B]\{q\}$, $[B]$ built from $\partial N_i/\partial x$ and $\partial N_i/\partial y$ (each node gives a 3×2 block)

$[B_i] = [\partial N_i/\partial x \; 0 \; ; \; 0 \; \partial N_i/\partial y \; ; \; \partial N_i/\partial y \; \partial N_i/\partial x]$

$[k] = t \int_{-1}^{1} \int_{-1}^{1} [B]^T [D] [B] \det[J] \, d\xi \, d\eta$

- The integrand is too complex for exact integration; Gauss quadrature is used (2×2 points for a 4-node element, 3×3 for 8 or 9 node).

7.5 Advantages of isoparametric elements

- Curved boundaries and irregular shapes handled accurately with one element.
- Same code for any shape; integration done in the fixed domain -1 to +1.
- Convergence criteria satisfied automatically (rigid body motion + constant strain), compatible along boundaries.
- Widely used in commercial software (ANSYS, ABAQUS, NASTRAN).

8. Isoparametric Elements for 2-D Analysis

- 4-node quadrilateral (Q4): bilinear, 8 dof.
- 8-node quadrilateral (Q8, serendipity): quadratic, curved edges, 16 dof.

- 9-node quadrilateral (Q9, Lagrange): 18 dof.
- Triangular isoparametric elements: 3-node and 6-node (area coordinates).
- Procedure summary:
- Step 1: Select the parent element and shape functions $N_i(\xi,\eta)$.
- Step 2: Compute $[J]$ and $\det[J]$ at the Gauss point.
- Step 3: Compute $[B]$ = function of $[J]^{-1}$ and $\partial N/\partial\xi$, $\partial N/\partial\eta$.
- Step 4: Evaluate $[B]^T[D][B]\det[J]$ at each Gauss point and sum with weights.

  $$[k] \approx t \sum_i \sum_j w_i w_j [B]^T[D][B] \det[J]$$
- Step 5: Assemble, apply boundary conditions, solve, compute stresses at Gauss points.

9. Formulation of CST Element – Summary

- Nodes 3, dof 6. Displacement linear: $u = a_1+a_2x+a_3y$.
- Shape functions $N_i = (\alpha_i + \beta_i x + \gamma_i y)/2A$.
- $[B]$ constant (3×6), strains constant.
- $[k] = t\,A\,[B]^T[D][B]$ (6×6).
- Stress $\{\sigma\} = [D][B]\{q\}$; load vectors from body force and edge traction.

Important questions: (1) derive $[B]$ and $[k]$ for CST, (2) compare CST and LST, (3) explain isoparametric formulation, (4) shape functions of 4-node and 8-node isoparametric elements, (5) a

numerical problem on CST.